

Optimal Requantization of Deep Grayscale Images and Lloyd–Max Quantization

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Abstract—The classic signal quantization problem was introduced by Lloyd. We formulate another, similar problem: The optimal mapping of digital fine grayscale images (such as 9–13 bits-per-pixel medical images) to a coarser scale (e.g., 8 bits per pixel on conventional computer monitors). While the former problem is defined basically in the real signal domain with smoothly distributed noise, the latter refers to an essentially digital domain. As we show in this paper, it is this difference that makes the classic quantization methods virtually inapplicable in typical cases of requantization of the already digitized images. We found experimentally that an algorithm based on dynamic programming provides significantly better results than Lloyd’s method.

Index Terms—Dynamic programming, grayscale image, Lloyd–Max quantization, medical images, requantization.

I. INTRODUCTION

USERS OF medical computer networks often experience difficulties when exchanging documents containing medical images. Such images normally require 9–13 bits per pixel, while conventional computers allow for 8 bits per pixel. Linear rounding-up [linear mapping (LM)] is usually unacceptable since important image details may be lost or hidden. In Section II, we formulate an optimal nonlinear mapping (OM) of a deep grayscale image to one in a coarser scale. The idea was to obtain a mapped image which would preserve as much details as possible, even at the expense of contrast distortions (it is possible to mitigate the issue of contrast distortion by displaying two views of the same image: LM and OM views).

The core step of OM is the least squares approximation of the original image by one having n (assume $n = 256$) different levels of gray. This problem is similar to Lloyd–Max signal quantization, which usually assumes continuously distributed noise. Our goal is to study if the Lloyd–Max approach is applicable to our problem, which assumes a different domain and a different data model. In Section III, we discuss this approach and explain essential differences between the requantization of digitized images and the Lloyd–Max quantization. Section IV refers to an alternative approach, based on dynamic programming (DP). The DP solution for optimal partitioning of the one-dimensional scale was introduced in [2]–[4]. In Section

V, we provide experimental research and comparison of Lloyd’s and the DP solutions. Based on this study, we conclude that in the optimal requantization of medical images, the DP algorithm provides substantially better results.

II. OPTIMAL IMAGE REQUANTIZATION: PROBLEM STATEMENT

Consider source image $A : \{\hat{a}_1, \dots, \hat{a}_M\}$ as an ordered set of M pixels, each pixel being defined by its coordinates and intensity value a_i . The order of pixels is assumed fixed, so image A can be identified by its intensity vector $\{a_1, \dots, a_M\}$ in M -dimensional space. Components a_i are integers within $0, N-1$, which defines a *source intensity scale*.

Let $B : \{b_1, \dots, b_M\}$ denote another image—a vector with components measured on the *target scale*, which is a set of integers in $0, n-1$, ($n < N$).

Given source image A , in order to find its best approximation on the target scale, we need to define the distance between images defined on different intensity scales.

Consider a set of n ordered real nonnegative parameters β^k , such that

$$0 \leq \beta^0 < \beta^1 < \dots < \beta^{n-1} \leq \max_{\{1 \leq i \leq M\}} (a_i) = a^{(\max)}. \quad (1)$$

Let D define distance between A and B , as follows:

$$D(A, B) = \min_{\{\beta^0, \dots, \beta^{n-1}\}} \sum_{i=1}^M (a_i - \beta^{b_i})^2. \quad (2)$$

Then, the optimally requantized image B_A is defined as

$$B_A = \arg \min_{\{B\}} D(A, B). \quad (3)$$

So, the minimization in (2) and (3) is provided by the best set $\{\beta^0, \dots, \beta^k, \dots, \beta^{n-1}\}$, as well as by the best choice of integer values $b_i \in 0, n-1, i \in 1, M$.

Note that we do not need set $\{\beta^k\}$ per se, since we are not going to build a codebook; it is just an intermediate means for getting an optimal B_A . This distinguishes our problem statement from the quantization/requantization in the literature.

III. LLOYD–MAX APPROACH TO IMAGE REQUANTIZATION

Lloyd showed that the approximation problem similar to (1)–(3) can be reduced to an optimal partitioning of the scale of signal values into n disjunctive semi-open intervals. This is a *key statement* in [1], and will be referred to thus below. The direct consequence is that for each k , the optimal β^k is the mean signal value in the k th interval.

The key statement is based on the following data model: 1) the observed signal values are real numbers; in our terms, $N = \infty$;

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2) infinite number of samples; in our case, the number of pixels, $M = \infty$ (specifically in [1], that follows from the Nyquist theorem); 3) the observed signal values differ from certain true values by additive and relatively smoothly distributed noise.

Although it was not said explicitly, an *ideal data model* would have been defined by a *continuous and strictly increasing cumulative distribution function (c.d.f.)*, $F(x)$. A good example might be a mixture of n normal distributions

$$F(x) = \sum_{k=0}^{n-1} P_k \int_{-\infty}^x \exp\left(-\frac{(z - \beta^k)^2}{2\sigma_k^2}\right) dz \quad (4)$$

where factors P_k account for normalization and probabilities of different quanta β^k .

In [1], the problem was stated for the c.d.f. $F(x)$ of general type; however, any deviation from the *continuity* and *strictly increasing behavior* was considered a nonessential and merely technical complication. This might be acceptable when the *ideal data model* assumption was true in the major part of domain. It is usually the case for the quantization of analog signals.

Practically, signal measurement always eventually results in conversion to digital form; so, we can assume a finite number of initial quanta N (speaking more accurately, a number of distinct initial quanta of nonzero probability). In order for the model to be close to the analog case, the following two inequalities should take place:

$$n \ll N \ll M \quad (5)$$

where the left one would make the interval include *many* initial quanta, so the probability of any particular value of the signal would be negligible; the right one would make a histogram close to the true distribution density.

In case of image requantization, e.g., if $n = 256$, $N < 1000$, $M = 256 * 256 = 65\,536$, at least the first inequality in (5) is not true. It is a combinatorial, rather than a regular optimization, problem. The c.d.f. has discontinuity in each integer $0 \leq x < N$, and intervals of constancy in between. There is no one “regular” (in terms of the *ideal model*) point at all in the entire domain. Hence, it is a different problem.

The first consequence of this difference is that the proof of the Lloyd’s *key statement* cannot be fully extrapolated to the digital image domain, because the special case of the signal value exactly at the boundary point (midway between the two adjacent quanta) is ignored. It was ignored in [1] and in later literature (e.g., [5, p. 176]). This is acceptable in the analog case, where any single value can be treated as one of probabilistic measure zero. In case of image requantization, endpoints with nonzero probability are quite possible. Therefore, it is important which of the two adjacent intervals the boundary intensity will be assigned to, according to optimal partitioning. Even more important is the question: Could part of the corresponding pixels belong to the left interval, while the rest belong to the right one? Had this split been possible, the *key statement* would not have been true in the digital domain.

The answer cannot be obtained using Lloyd’s reasoning; a different and independent proof of this statement in the digital domain is required. In [6], we showed in particular, that the optimality in (1)–(3) could never be reached with a “split endpoint”: If B is the optimal requantization of A , then for any

pair of pixel indices $i, j \in \overline{1, M}$, $(a_i \leq a_j) \Rightarrow (b_i \leq b_j)$; $(b_i < b_j) \Rightarrow (a_i < a_j)$; the last implication means that pixels with equal intensity values cannot fall in the different intervals.

The second consequence of the difference between the classic quantization and image requantization relates to the algorithms. While Lloyd’s *key statement*, with the above extension, holds true for the problem (1)–(3), both quantization methods in [1] face serious difficulties in the digital domain.

The basic idea of both heuristic solution methods in [1] is that the endpoint between adjacent intervals is always midway between the corresponding quanta

$$x^k = (\beta^{k-1} + \beta^k)/2. \quad (6)$$

Method I starts with random partition $\{x^1, \dots, x^n\}$. Each quantum β^k is calculated as an average of the signal values in the k th interval. Then, the endpoints of the intervals are adjusted according to (6), and the quanta are recalculated again. The iterations continue until a certain stopping criterion is met.

Method II starts with random value β^1 ; endpoint x^1 of the first interval is calculated to make β^1 an average in the first interval $(-\infty, x^1)$. Then, quantum β^2 is calculated to satisfy condition (6); endpoint x^2 of the second interval is calculated to make β^2 an average in (x^1, x^2) , and so forth. If the last quantum in this sequence β^n differs from the average signal value in the last interval by more than certain threshold ε , the process restarts from a new initial value β^1 . The process stops when the difference between β^n and the average in the last interval does not exceed ε .

Obviously, (6) is not sufficient for optimality, so the algorithms usually stop in the local minima. Moreover, it is not necessary, either, because optimal intervals may be separated not only by endpoints (6), but also by *separating intervals* (SIs) of nonzero length and probability 0. In our case, every SI has length of at least 1. Any point of such a SI, including midpoint defined in (6), may be treated as an endpoint between adjacent target intervals. Indefiniteness of the endpoints is inherent to our problem, while in [1], it is, rather, an exception.

Specifically, these SI (intervals of constancy of c.d.f. $F(x)$) are a real problem for Method II. If only a few SI exist, Lloyd proposed to add a few more minimization parameters. In our 8-bit scale example, there are exactly 255 intervals of constancy; it is hardly feasible to minimize by 255 additional parameters, so Method II is, in fact, inapplicable.

Unlike Method II, there are no obvious obstacles for using Method I, so its applicability to image requantization should also be studied experimentally. We describe our experiments and discuss the results in Section V.

IV. OPTIMAL REQUANTIZATION

As an alternative to Lloyd’s algorithms, a globally optimal image requantization (partitioning of the source scale for maximum homogeneity of intervals), based on DP, can be used. Although the algorithms in [2]–[4] are formally different, they are equivalent in terms of asymptotical computational complexity, which is $\sim nN^2$.

In the early days of quantization theory, DP was either unknown, or not feasible as a practical optimization method for the problem at hand. When applied to grayscale images, with modern computers, it takes just a few seconds for the DP algorithm to obtain an optimally requantized image.

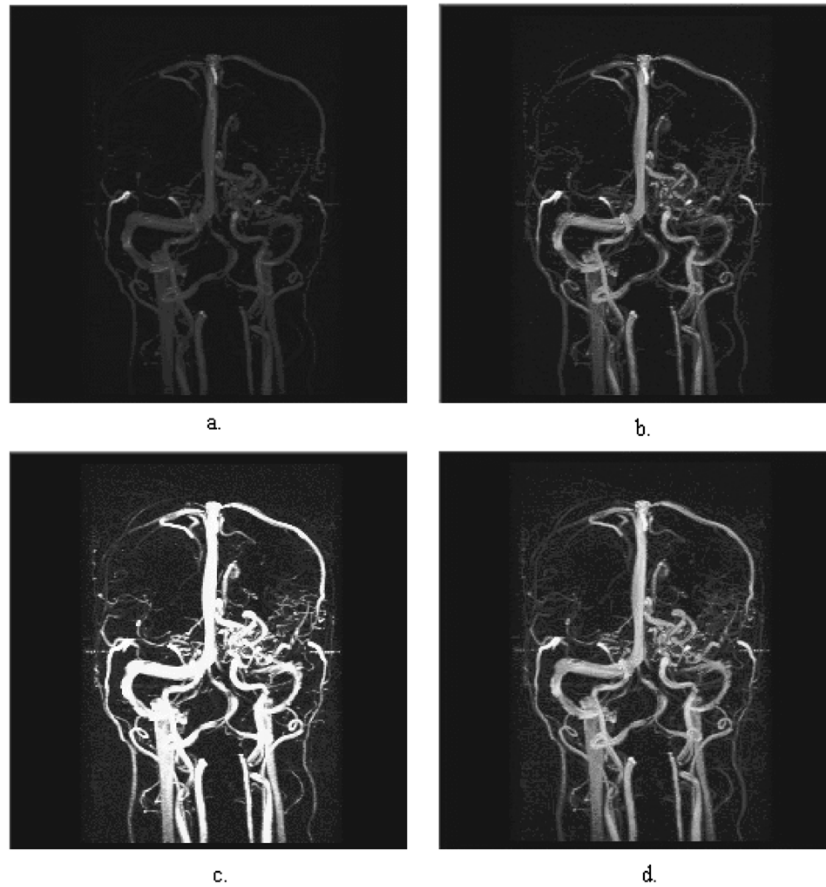


Fig. 1. Image MR-ANGIO, [7]. (a) Linearly mapped image; $D = 94\,376.01$. (b) Best image obtained by Method I, with randomly generated initial conditions; seed = 10; $D = 23\,364.43$. (c) Worst image obtained by Method I; trial 99, $D = 583\,217.56$. (d) Optimal requantization; $D = 6757.28$.

TABLE I
SUMMARY OF EXPERIMENTAL STUDY OF LLOYD'S METHOD I

Trial #	Initial partitioning	Iterations	Distance $D(A,B)$
1 through 98	Random	5 through 13	23364.43 through 57012.44
99	[0,1), [1,2), ... [254,255), [255,2048)	126	583217.56
100	Optimal	1	6757.28

In our experimental study, we used the algorithm provided in [6], which is close to [3].

V. EXPERIMENTAL RESULTS

We studied Lloyd's Method I experimentally. We used an 11-bit, 256×256 medical image [7], where only 632 intensity values out of 2048 had nonzero probability, in the range of 0–1365. The result of the linear mapping to 8-bit grayscale is presented in Fig. 1(a).

One hundred eleven trials were made in total using various initial conditions, in order to obtain 100 successful trials; 11 trials were prematurely terminated because an empty interval occurred during the iterations. An empty interval is a known complication of Method I, which, however, was viewed as a merely technical and virtually unlikely one [1]. We did not use modifications of the algorithm to handle these situations; we

simply terminated the trial and excluded it from the results in order to study a "pure" Method I.

The experimental results are summarized in Table I.

In trials 1–98, initial intervals were built at random, with uniform distribution and different seed values of the generator of pseudo-random numbers. Care was taken to make each initial interval nonempty (improper random values were skipped).

In trial 99, initial intervals were chosen as follows: first 255 intervals contained single values 0,1,2,...,254, respectively; the last interval contained values 255 through the maximum, 1365.

In trial 100, for initial partitioning, we used an *optimal* partitioning, obtained by the DP algorithm.

The following observations have been made.

- 1) All of the obtained 100 solutions were different.
- 2) Only trial 100 resulted in the optimal solution. (Note: Had it happened a boundary intensity value assigned to

incorrect interval, Method I would have pulled away from the optimal starting partition and eventually would have stopped at a nonoptimal one; that did not happen).

- 3) While the optimal value of $D(A, B)$ was 6757.28, the range of values obtained in trials 1–98 was 23 364.43 through 57 012.44. In trial 99, it was 583 217.56. For the linear mapping, $D = 94 376.01$. In all trials, the greater initial D (corresponding to initial partition), the greater the final value of criterion D .
- 4) All of the images obtained in trials 1–99 are visually worse than the optimal image Fig. 1(d). By “worse,” we mean that less image detail are visually distinguishable in the picture. As a rule, the greater $D(A, B)$ value, the worse the picture visually.

Fig. 1(b) presents the best image, obtained in trials 1–99; Fig. 1(c) presents the worst of the images obtained in trials 1–99 (no details can be distinguished in the white area.)

One can see that the OM image Fig. 1(d) contains image detail that is invisible even in the best image Fig. 1(b) obtained by Lloyd’s Method I in trials 1–99.

The reason for this behavior is that with $n = 256$, $N = 632$, condition (5) is not met. There is no sufficient leeway for Method I to improve initial partitioning. The algorithm usually stops after just a few iterations (6–8 in most of our experiments). That does not happen in the real [or digital, provided (5)] domain, where fine endpoint positioning may allow for a long iteration process, thus providing a potentially better solution.

The following example illustrates this in more detail. Assume that before the current iteration, the endpoint between the adjacent intervals was 5.1; after recalculation of $\{\beta^k\}$, followed by recalculation of endpoints according to (6), we obtain a new endpoint 5.9, while all the other endpoints did not change. After this iteration, the algorithm will stop, because the interval between 5.0 and 6.0 is its “zone of insensitivity.” Repositioning of the endpoint from 5.1 to 5.9 will not cause further changes and a continuation of iterations.

As a result, the obtained images are highly dependent on the initial conditions of Method I.

VI. CONCLUSION

We introduced a formal statement of the image requantization problem, which is close to the Lloyd–Max signal quantization. However, there exists a substantial difference between the two.

Lloyd–Max quantization is based on the idea that endpoints of optimal intervals are located midway between the corresponding quanta (6). Generally, it is neither necessary nor sufficient for optimality. It works in the real domain, in the case of distribution without singularities and intervals of constancy of c.d.f., or in the digital domain (5).

When we switch to image requantization, with nothing but discontinuity points and intervals of constancy being a c.d.f. domain, and with (5) being false, the *key statement* (the equivalence of the signal approximation problem and the optimal partitioning of signal scale) holds true, but it does not follow from the simplified reasoning in [1]. An independent proof in [6] provides an accurate treatment of all cases, including those having probability 0 in the real domain.

As for the two classic heuristic solution methods for suboptimal partitioning [1], the first one shows rather poor behavior

in the case of image requantization; the second one is, in fact, inapplicable.

A globally optimal solution, based on DP, seems to be a real alternative for image requantization.

There also exist numerous suboptimal heuristic algorithms, which may work on images significantly better than Lloyd’s quantization; that was beyond the scope of this paper.

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